

Lesson 20: Solving Trigonometric Equations — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Solve $\sin \theta = 1$, 0° – 360° .

2. Solve $\cos \theta = 0$, 0° – 360° .

3. Solve $\tan \theta = 0$, 0° – 360° .

4. Solve $\sin \theta = 0$, 0° – 360° .

5. Solve $\cos \theta = 1$, 0° – 360° .

6. $\sin^{-1}(0.5) = ?$

Section 2 — Core practice

7. Solve $\cos \theta = 0.5$, 0° – 360° .

8. Solve $\tan \theta = 1$, 0° – 360° .

9. Solve $\sin \theta = \sqrt{2}/2$, 0° – 360° .

10. Solve $\cos \theta = -0.5$, 0° – 360° .

11. Solve $\sin \theta = -0.5$, 0° – 360° .

12. Solve $\tan \theta = \sqrt{3}$, 0° - 360° .

Section 3 — Challenge & exam-style

13. Solve $2 \sin \theta = 1$, 0° - 360° .

14. Solve $2 \cos \theta + 1 = 0$, 0° - 360° .

15. Solve $\sqrt{2} \sin \theta = 1$, 0° - 360° .

16. Solve $\tan \theta - 1 = 0$, 0° - 360° .